4. AI Requirements

The application requires a Gemini API connection, a secure API key stored on the server, a prompt-generation component and response handling logic.

Tabulation

Feature	Purpose
Profile	Capture user preferences
Activity	Choose preferred activity
Time	Specify available duration
Generate	Request AI plan

Coding / Technical Example

```
# FitBuddy technical flow
user_input = {
    "experience": "beginner",
    "activity": "walking",
    "time": "30 minutes"
}

prompt = f"Create a general wellness plan for {user_input}"
# Send prompt to Gemini from the secure backend.
# Display the generated response in the UI.
```

Picture / Diagram

```
PROFILE FORM → VALIDATION → GENERATE BUTTON → PLAN
```

5. Non-Functional Requirements

The application should be easy to use, responsive, maintainable, reasonably fast, secure in handling API credentials and clear about the limits of AI-generated wellness information.

Tabulation

AI Element	Role
Prompt	Defines task and constraints
Gemini	Generates natural-language output
Response	Returned AI content
Formatter	Creates readable result

Coding / Technical Example

```
# FitBuddy technical flow
user_input = {
    "experience": "beginner",
    "activity": "walking",
    "time": "30 minutes"
}

prompt = f"Create a general wellness plan for {user_input}"
# Send prompt to Gemini from the secure backend.
# Display the generated response in the UI.
```

Picture / Diagram

```
USER DATA → PROMPT → GEMINI → AI RESPONSE → FORMATTER
```

6. Hardware / Software Requirements

Hardware: a computer or mobile device with internet access. Software: Python, Flask or another backend framework, HTML/CSS/JavaScript, Gemini SDK/API access and optionally SQLite/MySQL.

Tabulation

Stage	Action
Input	Collect information
Process	Validate and generate prompt
AI	Send request to Gemini
Output	Display plan

Coding / Technical Example

```
# FitBuddy technical flow
user_input = {
    "experience": "beginner",
    "activity": "walking",
    "time": "30 minutes"
}

prompt = f"Create a general wellness plan for {user_input}"
# Send prompt to Gemini from the secure backend.
# Display the generated response in the UI.
```

Picture / Diagram

```
INPUT → PROCESS → AI → OUTPUT
```